

EXPERIMENT 20 – Stock Analysis

Question

Analyze stock prices using line chart, percentage-change bar chart, and table.

Dataset

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

The supplied Stock Prices dataset contains three stocks over three dates.

R Code

```
# Experiment 20 - Stock Analysis

date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03"))

stock_a <- c(100, 105, 110)
stock_b <- c(150, 152, 148)
stock_c <- c(120, 118, 122)

data <- data.frame(date, stock_a, stock_b, stock_c)

# Question 1 - Line Chart

plot(date, stock_a,
      type = "o",
      ylim = range(c(stock_a, stock_b, stock_c)),
      main = "Stock Price Comparison",
      xlab = "Date",
      ylab = "Stock Price")
```

```

lines(date, stock_b, type = "o")
lines(date, stock_c, type = "o")
legend("topleft",
      legend = c("Stock A", "Stock B", "Stock C"),
      lty = 1,
      pch = 1)

```

Question 2 - Daily Percentage Change

```

percentage_change <- c(NA, diff(stock_a) / stock_a[-length(stock_a)] * 100)
barplot(percentage_change[-1],
      names.arg = format(date[-1], "%d-%b"),
      main = "Daily Percentage Change - Stock A",
      xlab = "Date",
      ylab = "Percentage Change (%)",
      col = "lightgreen")

```

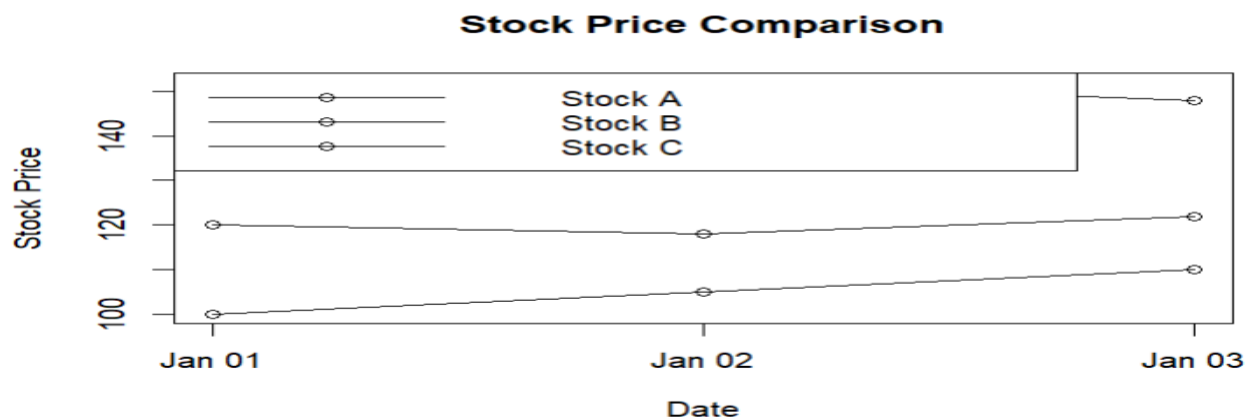
Question 3 - Table

```
print(data)
```

Question 1

Create a line chart to visualize stock prices of Stock A, Stock B, and Stock C.

Output 1

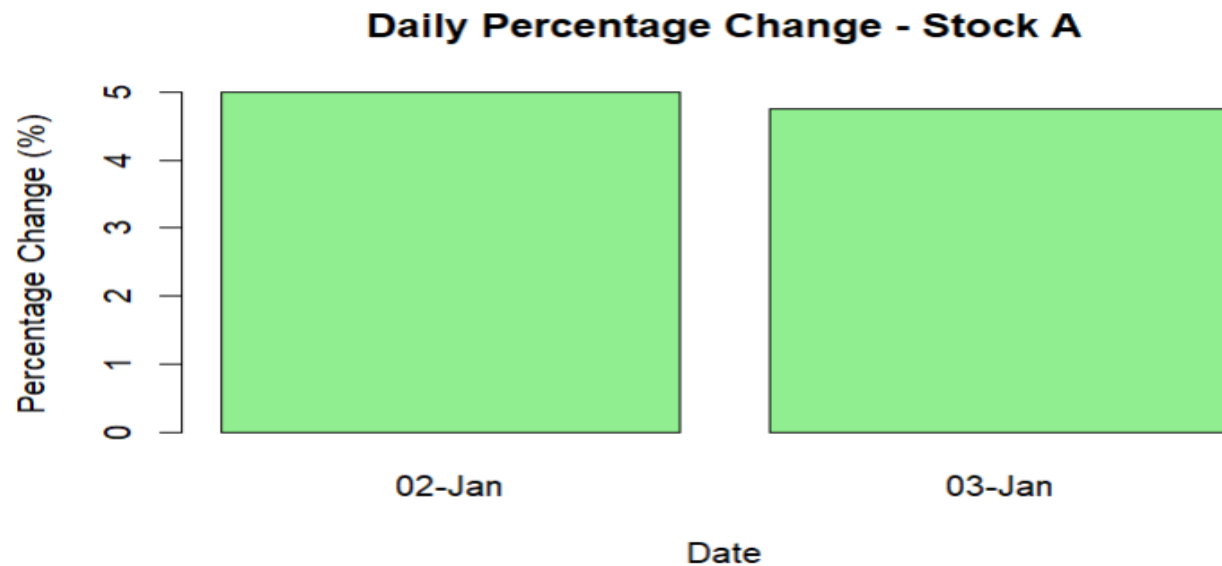


Stock A increases from 100 to 110, Stock B fluctuates from 150 to 148, and Stock C fluctuates from 120 to 122.

Question 2

Generate a bar chart showing daily percentage changes in Stock A.

Output 2



Stock A increases by approximately:

2023-01-02 = 5%

2023-01-03 = 4.76%

Question 3

Display stock price data.

Output 3

```
> # Question 3 - Table
> print(data)
      date stock_a stock_b stock_c
1 2023-01-01     100     150     120
2 2023-01-02     105     152     118
3 2023-01-03     110     148     122
```